Exercise Sheet

Task 1 – Euclidean Algorithm

Below, you will find the pseudocode for the Euclidean algorithm, which is used to compute the greatest common divisor (GCD) of two numbers greater than zero.

a) What is the GCD of 20 and 6?

$gcd(20, 6) = 2$

b) Translate the pseudocode-snippet into assembly code.

```
gcd(a,b):
   while b != 0
     t := b
     b :=a mod b
     a := t
   return a
```

Presuming that a is initialized in t0 and b is initialized in t1:

```
loop:
   add t2, t1, zero
   rem t1, t0, t1
   add t0, t2, zero
   bne t1, zero, loop
```

Task 2 – Negabinary Numbers

The negabinary representation of a given number x is its representation to the base of -2. Given the coefficients $a_n, a_{n-1}, , a_1, a_0$ the number can be represented using this formula:

$$x = \sum_{i=0}^{n} a_i(-2)^i = \ + a_2(-2)^2 + a_1(-2)^1 + a_0(-2)^0$$

(a) Convert $(101101)_{-2}$ and $(011101)_{-2}$ to their decimal representation.

$(101101)_{-2} = (-35)_{10}$
$(011101)_{-2} = (13)_{10}$

(b) Convert $(12)_{10}$ and $(-23)_{10}$ to their negabinary representation using 6 bits to represent the number.

$(12)_{10} = (011100)_{-2}$
$(-23)_{10} = (111001)_{-2}$

(c) List all values in negabinary and decimal representation that can be displayed with n = 2. In your own words, find a general rule to describe the range of negabinary numbers for different n.

Negabinary	Decimal
000	0
001	1
010	-2
011	-1
100	4
101	5
110	2
111	3

The maximum number that can be displayed in negabinary representation of n bits is the sum of all bits with an even exponent, the minimum number is the sum of all bits with an uneven exponent.

Task 3 – BIOS/UEFI

When looking at different types of memory you already came across BIOS and UEFI, research these two to answer the following questions:

- What are the tasks that these firmware interfaces perform?
- What are the key differences between those two?
- Where are those saved?

- Can they be updated, what difficulties might arise?

- Tasks:
  - Test all hardware components for functionality, and initialize them to bring the PC into a functional state and then give control to an operating system.
  - Enable hardware configuration: clock speed, temperature limits for emergency shutdown, enable or disable hardware components, interfaces or functions.
  - Boot priorities
  - Save date and time
- Key differences:
  - BIOS has to be navigated with a keyboard while UEFI supports mouse inputs
  - In contrast to BIOS, UEFI has a standard network driver, so that firmware updates can be downloaded without a full system start
  - UEFI supports up to 128 hard disk partitions, while BIOS only supports up to 4
  - UEFI has extended security measures like secure boot and TPM
- Chip:
  - E.g. EEPROM, EPROM, Flash memory...
- Update:
  - Years ago the BIOS could not be updated, and the whole chip had to be replaced instead
  - Nowadays the chips used are reprogrammable, but only a couple of hundred times, so the UEFI should not be updated too often
  - If a BIOS/UEFI update goes wrong, the mainboard can become unusable